# The Poole Local-Geometry Program: Hysteretic Surface Selection, Defect Calculus, and Channel Spectra in a Three-Dimensional Cellular Automaton

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June 2026

**Abstract**

This manuscript develops a self-contained mathematical account of a three-dimensional cellular automaton governed by local support windows on the 26-neighbor Moore lattice. The base binary rule is

$$P(x)_i = \mathbf{1}\{5 \leq N_i(x) \leq 7 + 2x(i)\},$$

which is exactly the birth-survival rule B5–7/S5–9. The state-dependent term $2x(i)$ is a local hysteretic widening of the survival window. From this single rule one obtains stable one-cell coordinate sheets, rejection of ideal points, lines, and filled bulk, exact first-step planar defect laws, pore-gas fixed subfamilies, rectangle response formulas, channel spectra, work identities, selected non-planar first-response formulas, and a channel-resolved geometry transform for measured binary fields. A signed extension with states $-1, 0, +1$ is then formulated as a local opposite-support interface model; its morphology scores are defined as descriptors only, not as physical energy or thrust laws. The final bridge theorem identifies the planar defect selector with the channel-resolved PooleQ/P spectrum. All physical, continuum, hardware, and unknown-quantum-state claims are excluded from the established results.

## Contents

# 1 Purpose and boundary

The purpose of this paper is narrow: it gives a finite local-geometry theory for a deterministic cellular automaton and for channel-resolved transforms derived from the same local support counts. The results below are statements about finite lattices, local neighbor counts, channel decompositions, and first-step or explicitly stated finite-step identities.

The paper does not establish a physical theory of spacetime, gravity, antimatter control, propulsion, energy extraction, quantum-state cloning, or hardware implementation. When quantum or p-bit language appears, it is used only for measured classical binary fields or probability variables. The standard no-cloning boundary is respected: an unknown quantum state cannot be copied by the constructions discussed here [1, 2].

The manuscript separates five lanes:

1. theorem-level statements proved from finite local combinatorics;
2. definition-level objects introduced for later use;
3. formula-level identities valid under stated hypotheses;
4. model-tagged variants that are not the raw rule;
5. open problems and conjectures.

No numerical artifact is needed to understand or apply the mathematical statements in this version.

# 2 The finite Poole lattice

**Definition 2.1** (Periodic cubic lattice)**.** *Fix $L \in \mathbb{N}$. The finite periodic cubic lattice is*

$$\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^3.$$

*A binary configuration is a map $x : \Lambda_L \to \{0, 1\}$. The value $x(i) = 1$ means that site $i$ is active.*

**Definition 2.2** (Moore neighborhood and support count)**.** *The radius-one 26-neighbor Moore neighborhood is*

$$\mathcal{N}_{26} = \{-1, 0, 1\}^3 \setminus \{(0, 0, 0)\}.$$

*For $i \in \Lambda_L$, the active support count is*

$$N_i(x) = \sum_{\delta \in \mathcal{N}_{26}} x(i + delta),$$

*with addition performed modulo $L$ in each coordinate.*

**Definition 2.3** (Base Poole map)**.** *The base Poole map $P : \{0, 1\}^{\Lambda_L} \to \{0, 1\}^{\Lambda_L}$ is*

$$P(x)_i = \mathbf{1}\{5 \le N_i(x) \le 7 + 2x(i)\}. \tag{1}$$

*The dynamics is $x_{t+1} = P(x_t)$.*

**Theorem 2.4** (Compact-rule equivalence)**.** *Equation* (1) *is exactly the birth-survival rule B5–7/S5–9: an inactive site is born iff $N_i \in \{5, 6, 7\}$, and an active site survives iff $N_i \in \{5, 6, 7, 8, 9\}$.*

*Proof.* If $x(i) = 0$, then $7 + 2x(i) = 7$, so (1) accepts exactly $5 \le N_i \le 7$. If $x(i) = 1$, then $7 + 2x(i) = 9$, so it accepts exactly $5 \le N_i \le 9$. □

**Remark 2.5** (Local hysteresis)**.** *The map is deterministic and Markovian in the full configuration. It is locally hysteretic because the current center state changes the upper support capacity by two. Active structure receives a two-neighbor persistence allowance.*

# 3 Equivalent normal forms

## 3.1 Birth-survival form

The rule can be written

$$P(x)_i = (1 - x(i))\mathbf{1}\{N_i \in \{5, 6, 7\}\} + x(i)\mathbf{1}\{N_i \in \{5, 6, 7, 8, 9\}\}.$$

## 3.2 Capacity form

Define the state-dependent capacity

$$C_i(x) = 7 + 2x(i).$$

Then

$$P(x)_i = \mathbf{1}\{5 \le N_i(x) \le C_i(x)\}.$$

## 3.3 Deficit-excess form

Define

$$D_i(x) = (5 - N_i(x))_+, \qquad E_i(x) = (N_i(x) - C_i(x))_+,$$

where $a_+ = \max\{a, 0\}$. The signed support strain is

$$\psi_i(x) = E_i(x) - D_i(x).$$

**Proposition 3.1** (Zero-strain selection)**.** *For the base rule,*

$$P(x)_i = 1 \quad \Longleftrightarrow \quad D_i(x) = E_i(x) = 0 \quad \Longleftrightarrow \quad \psi_i(x) = 0.$$

*Proof.* The equality $D_i = E_i = 0$ is exactly the condition $5 \le N_i \le C_i$. Substituting $C_i = 7 + 2x(i)$ gives (1). Since $D_i, E_i \ge 0$, $\psi_i = 0$ is equivalent to $D_i = E_i = 0$ only in the accepted window; outside the window one of the two nonnegative terms is strictly positive. The intended use of $\psi_i$ is therefore as a signed diagnostic together with the deficit-excess pair. □

## 3.4 Channel form

For $k = 0, 1, \ldots, 26$, define

$$B_k(i; x) = (1 - x(i))\mathbf{1}\{N_i(x) = k\}, \qquad S_k(i; x) = x(i)\mathbf{1}\{N_i(x) = k\}.$$

Then

$$P(x)_i = \sum_{k=5}^{7} B_k(i; x) + \sum_{k=5}^{9} S_k(i; x).$$

The accepted birth channels are $B_5, B_6, B_7$ and the accepted survival channels are $S_5, S_6, S_7, S_8, S_9$.

# 4 Basic structural facts

**Proposition 4.1** (Cubic symmetry and translation equivariance)**.** *The map $P$ commutes with translations of $\Lambda_L$, coordinate permutations, and coordinate sign flips.*

*Proof.* Each listed operation permutes the 26 Moore offsets. Therefore it preserves the multiset of neighbor values in every local block and hence preserves $N_i(x)$ and the update condition. □

**Proposition 4.2** (Finite propagation)**.** *If two configurations agree on the radius-$t$ Moore ball around $i$, then their states at $i$ agree after $t$ updates. Influence speed is at most one lattice unit per update in Chebyshev distance.*

*Proof.* One update at $i$ depends only on the radius-one Moore block around $i$. Iterating this dependency gives the radius-$t$ dependency after $t$ updates. □

**Proposition 4.3** (Eventual periodicity)**.** *For fixed $L$, every trajectory under $P$ is eventually periodic. The preperiod plus period is at most $2^{L^3}$.*

*Proof.* There are $2^{L^3}$ configurations. A deterministic map on a finite set must eventually revisit a configuration. After a repeat, determinism forces periodicity. □

**Proposition 4.4** (Non-monotonicity)**.** *The map $P$ is not monotone with respect to coordinatewise order.*

*Proof.* Choose an inactive site $i$. Let $x$ contain exactly five active Moore neighbors of $i$. Let $y \geq x$ be obtained by adding three more active neighbors of $i$, with $x(i) = y(i) = 0$. Then $N_i(x) = 5$ and $P(x)_i = 1$, while $N_i(y) = 8$ and $P(y)_i = 0$ because inactive birth is allowed only through $5, 6, 7$. □

**Proposition 4.5** (Non-reversibility)**.** *For $L \geq 3$, $P$ is not injective and therefore is not reversible.*

*Proof.* The all-zero configuration maps to itself. The all-one configuration has $N_i = 26$ at every site, so every active site is over-supported and dies. Thus the all-one configuration also maps to the all-zero configuration. □

# 5 Fixed points and active graph constraints

**Definition 5.1** (Active adjacency graph)**.** *For a configuration $x$, let $G_A(x)$ be the graph whose vertices are active sites and whose edges connect active sites at Moore-neighbor distance.*

**Proposition 5.2** (Fixed-point constraints)**.** *A configuration $x$ is a fixed point iff every site satisfies*

$$x(i) = 1 \Rightarrow 5 \leq N_i(x) \leq 9, \qquad x(i) = 0 \Rightarrow N_i(x) \notin \{5, 6, 7\}.$$

*Proof.* An active site must survive and an inactive site must avoid birth. These are exactly the two displayed conditions. □

**Corollary 5.3** (Active-degree bounds)**.** *If $x$ is a nonzero fixed point and $A = \{i : x(i) = 1\}$, then every active vertex of $G_A(x)$ has active Moore degree between $5$ and $9$. Hence*

$$5 \leq \frac{2|E(G_A)|}{|A|} \leq 9.$$

## 6 Hysteretic surface selection

**Definition 6.1** (Coordinate-flat local pattern)**.** *A configuration is locally a coordinate $r$-flat through a site $i$ if, inside the $3 \times 3 \times 3$ Moore block centered at $i$, the active sites are exactly the sites in an $r$-dimensional coordinate subspace through $i$.*

**Lemma 6.2** (Flat local degree)**.** *If an active site is locally a coordinate $r$-flat, then*

$$N_i = 3^r - 1.$$

*Thus the local degrees for a point, line, sheet, and filled block are $0, 2, 8, 26$.*

*Proof.* The centered coordinate $r$-flat contributes $3^r$ active sites inside the $3 \times 3 \times 3$ block, one of which is the center. The remaining $3^r - 1$ are active neighbors. □

**Theorem 6.3** (Local surface selector)**.** *Among ideal coordinate-flat local bulk types in three dimensions, the survival window $5 \leq N_i \leq 9$ preserves exactly the locally two-dimensional sheet type.*

*Proof.* The possible coordinate-flat local degrees are $0, 2, 8, 26$. Only 8 lies in $[5, 9]$. Therefore points and lines are under-supported, filled bulk is over-supported, and sheets survive. □

**Theorem 6.4** (Stable active coordinate plane)**.** *Assume $L \geq 3$. Fix a coordinate direction and a level $c$. Let*

$$H_c = \{(a, b, c) : a, b \in \mathbb{Z}/L\mathbb{Z}\} \subset \Lambda_L,$$

*and let $x = \mathbf{1}_{H_c}$. Then $P(x) = x$.*

*Proof.* An active site in the sheet sees the other eight sites in its in-plane $3 \times 3$ block, so $N = 8$ and the site survives. A site in either adjacent normal layer sees the full $3 \times 3$ footprint of the sheet, so $N = 9$, which is above the inactive birth window. Sites farther away see $N = 0$. Thus the plane is unchanged. □

**Corollary 6.5** (The $+2$ term is sheet memory)**.** *If the memory allowance is removed and the survival window is reduced to $5, 6, 7$, then a full one-cell active plane dies in one step, because its active sites have $N = 8$.*

**Proposition 6.6** (Filled slabs are rejected)**.** *Let $S$ be a union of at least two consecutive full coordinate sheets. Then no active site in $S$ survives under the base rule, and no inactive site is born from the slab. Hence $P(\mathbf{1}_S) = 0$.*

*Proof.* An active site in such a slab sees eight active neighbors in its own sheet and at least nine in an adjacent sheet, so $N \geq 17 > 9$. It dies by over-support. An inactive site outside the slab sees 0, 9, or 18 full-sheet neighbors, none of which lies in $\{5, 6, 7\}$. □

## 7 Planar defect calculus

The exact planar defect calculus begins by removing a mask from a stable sheet.

**Definition 7.1** (Defective sheet)**.** *Identify the coordinate sheet with the two-dimensional torus*

$$\mathbb{T}_L = (\mathbb{Z}/L\mathbb{Z})^2.$$

*Let $D \subset \mathbb{T}_L$ be a finite set of missing sheet sites. The corresponding three-dimensional configuration is*

$$x_D(u, c) = \mathbf{1}\{u \notin D\}, \qquad x_D(u, z) = 0 \quad (z \neq c).$$

**Definition 7.2** (Open and closed defect counts)**.** *For $u \in \mathbb{T}_L$, define*

$$q_D(u) = \sum_{\eta \in \{-1,0,1\}^2 \setminus \{0\}} \mathbf{1}\{u + \eta \in D\},$$

$$r_D(u) = \sum_{\eta \in \{-1,0,1\}^2} \mathbf{1}\{u + \eta \in D\}.$$

*Thus $q_D$ counts defects in the open planar eight-neighborhood and $r_D$ counts defects in the closed $3 \times 3$ footprint.*

**Theorem 7.3** (One-step planar defect calculus)**.** *For the base Poole rule, the first update of $x_D$ is determined by $q_D$ and $r_D$ as follows.*

1. *An active sheet cell $u \notin D$ survives iff $q_D(u) \leq 3$ and dies iff $q_D(u) \geq 4$.*
2. *An inactive in-plane defect cell $u \in D$ is born in the original sheet plane iff $1 \leq q_D(u) \leq 3$.*
3. *An inactive cell one normal layer above or below the sheet, at $(u, c+1)$ or $(u, c-1)$, is born iff $2 \leq r_D(u) \leq 4$.*
4. *All layers with $|z - c| \geq 2$ remain inactive after the first update.*

*Proof.* All active sites initially lie in the sheet. If $u \notin D$, then the active neighbor count in the sheet plane is

$$N(u, c) = 8 - q_D(u).$$

Survival requires $5 \leq 8 - q_D(u) \leq 9$, which is equivalent to $q_D(u) \leq 3$. If $u \in D$, the same count applies but the site is inactive, so birth requires $5 \leq 8 - q_D(u) \leq 7$, equivalent to $1 \leq q_D(u) \leq 3$. A site one layer above or below the sheet sees the closed $3 \times 3$ sheet footprint, hence

$$N(u, c \pm 1) = 9 - r_D(u).$$

Birth requires $5 \leq 9 - r_D(u) \leq 7$, equivalent to $2 \leq r_D(u) \leq 4$. Sites at normal distance at least two see no active neighbors at the first update. □

**Corollary 7.4** (Stable isolated pore gas)**.** *If every closed $3 \times 3$ footprint contains at most one defect site, equivalently $r_D(u) \leq 1$ for all $u$, then $x_D$ is a fixed point. In particular, a single one-cell pore in an otherwise complete sheet is exactly stable.*

*Proof.* If $r_D \leq 1$, no normal-layer birth can satisfy $2 \leq r_D \leq 4$. Each defect site has $q_D = 0$, so no in-plane rebirth occurs. Every active sheet site has $q_D \leq 1 \leq 3$, so all active sheet sites survive. □

**Proposition 7.5** (Exact no-event criterion)**.** *The first update of $x_D$ has no births and no deaths iff the following three local conditions hold:*

$$q_D(u) \leq 3 \quad (u \notin D),$$

$$q_D(u) \notin \{1, 2, 3\} \quad (u \in D),$$

$$r_D(u) \notin \{2, 3, 4\} \quad (u \in \mathbb{T}_L).$$

*Proof.* The three conditions are exactly the negations of the three possible first-step event classes in Theorem 7.3: active sheet deaths, in-plane births, and normal-layer births. □

**Conjecture 7.6** (No-event converse)**.** *For localized planar defect masks in a sufficiently large periodic sheet, the no-event criterion is equivalent to the isolated pore-gas condition $r_D \leq 1$ everywhere. The isolated pore-gas direction is Corollary 7.4; the converse is left as an open finite-combinatorial classification problem.*

# 8 Planar examples and operators

## 8.1 Single pore

For $|D| = 1$, every closed footprint contains at most one defect. Corollary 7.4 applies: the pore is stable and emits no normal halo.

## 8.2 Separated pore gas

If pairwise Chebyshev distance between distinct defect sites is at least two in the sheet, then $r_D \leq 1$ everywhere. The entire defect mask is fixed.

## 8.3 Cracks and line sources

A contiguous line of missing cells violates the isolated pore-gas condition. Theorem 7.3 shows that normal-layer births occur above and below sites whose closed $3 \times 3$ defect footprints contain two, three, or four missing sites. Thus cracks are local sources for normal-layer response.

# 9 Rectangle puncture laws

Let $D$ be an axis-aligned $a \times b$ rectangle of missing cells in the sheet, with $a, b \geq 2$ and no wrap-around self-interaction. Let

$$P_\partial = 2a + 2b$$

be the rectangle perimeter in the planar square lattice.

**Theorem 9.1** (Rectangle zero-death first response)**.** *For an $a \times b$ rectangle defect with $a, b \geq 2$, no active sheet cell dies at the first update.*

*Proof.* An active sheet site can die only if it has at least four missing cells in its open $3 \times 3$ planar neighborhood. For a rectangular hole, an active site adjacent to the boundary can see at most three missing cells in its open footprint: along a side it sees three, near a corner it sees one or two depending on position, and away from the boundary it sees none. Hence $q_D(u) \leq 3$ for every active sheet site. □

**Theorem 9.2** (Raw rectangle birth count)**.** *For an $a \times b$ rectangle defect with $a, b \geq 2$, the total first-step raw birth count is*

$$B_0(a, b) = 4a + 4b + 12 = 2P_\partial + 12.$$

*Proof.* By Theorem 7.3, births are determined by the planar open count $q_D$ for in-plane defect cells and the closed count $r_D$ for the two normal layers. The accepted normal condition $2 \leq r_D \leq 4$ selects a one-cell thick collar around the rectangle footprint in each normal layer, with corner corrections. The accepted in-plane condition $1 \leq q_D \leq 3$ selects the non-isolated boundary part inside the rectangle. Counting the side contributions and the four corner neighborhoods yields $4a + 4b + 12$. Equivalently, the channel count in Theorem 9.3 sums to the same expression. □

**Theorem 9.3** (Rectangle channel spectrum)**.** *For an $a \times b$ rectangle defect with $a, b \geq 2$, the first-step birth-channel spectrum is*

$$(N_5, N_6, N_7) = (12,\ 4a + 4b - 16,\ 16).$$

*Thus $N_5 + N_6 + N_7 = 4a + 4b + 12$.*

*Proof.* The planar selector gives the birth channel by the active neighbor count. In-plane births use $N = 8 - q_D$ and normal births use $N = 9 - r_D$. The $B_5$ sites are the twelve deep-corner and corner-adjacent footprint cases whose accepted support count is five. The $B_7$ sites are the sixteen shallow corner-edge cases at the high birth edge. The side collars contribute the remaining $4a + 4b - 16$ sites at channel $B_6$. Summing the three groups gives Theorem 9.2. □

**Remark 9.4** (Rectangle law status)**.** *The rectangle formulas are raw-rule first-step formulas. They do not assert a closed multi-step relaxation law.*

## 10 Prime-sharpened model as a tagged variant

**Definition 10.1** (Raw and prime-sharpened tags)**.** *The tag PM0 denotes the raw base map. The tag PM$\varphi$ denotes the model-tagged first-step rectangle response in which the raw $B_7$ group is removed from the rectangle birth spectrum while $B_5$ and $B_6$ are retained. PM$\varphi$ is not the base rule.*

**Proposition 10.2** (Prime-sharpened rectangle count)**.** *For an $a \times b$ rectangle defect with $a, b \geq 2$, the PM$\varphi$ rectangle response count is*

$$B_\varphi(a, b) = N_5 + N_6 = 12 + (4a + 4b - 16) = 4a + 4b - 4.$$

*Proof.* Substitute the rectangle channel spectrum from Theorem 9.3 and delete the $N_7 = 16$ group by definition of PM$\varphi$. □

## 11 First-step work identities

Let $I_5, I_6, I_7 \geq 0$ be chosen channel weights for births in channels $B_5, B_6, B_7$.

**Definition 11.1** (First-step birth work)**.** *The weighted first-step birth work is*

$$W_B(x) = \sum_{k=5}^{7} I_k B_k(x),$$

*where $B_k(x)$ counts births in channel $k$.*

**Theorem 11.2** (Rectangle work law)**.** *For an $a \times b$ rectangle defect with $a, b \geq 2$,*

$$W_B(a, b) = 12I_5 + (4a + 4b - 16)I_6 + 16I_7.$$

*For the PM$\varphi$ tagged response,*

$$W_{B,\varphi}(a, b) = 12I_5 + (4a + 4b - 16)I_6.$$

*Proof.* Multiply the channel counts in Theorem 9.3 by their weights and sum. The PM$\varphi$ identity deletes the $B_7$ contribution by definition. □

## 12 Line-hole edge law

A one-cell-wide line hole is a boundary case not covered by the $a, b \geq 2$ rectangle law. If $D$ is an $n \times 1$ missing rectangle with no wrap-around interaction, then the planar selector still applies, but the footprint intersections differ because the two long sides overlap in the closed $3 \times 3$ neighborhoods. The line-hole family should therefore be treated as an edge law rather than as a rectangle specialization.

**Theorem 12.1** (Line-hole first response)**.** *Let $D$ be an $n \times 1$ line-hole defect in an otherwise stable coordinate sheet, with no wrap-around self-interaction. If $n = 1$, then $D$ is a stable isolated pore and no first-step events occur. If $n \geq 2$, then no active sheet cell dies, the first-step birth count is*

$$B_0(n) = 7n,$$

*and the birth-channel spectrum is*

$$(N_5, N_6, N_7) = (0, 7n - 14, 14).$$

*Thus the one-cell line family is an edge law, not the $b = 1$ specialization of the rectangle formula $4a + 4b + 12$.*

*Proof.* For $n = 1$, every closed $3 \times 3$ defect footprint contains at most one missing site, so Corollary 7.4 applies. Assume $n \geq 2$. Along the missing line, an endpoint defect has one other defect in its open footprint, and an interior defect has two. Hence every missing in-plane line site is reborn: the two endpoints are channel $B_7$, and the $n - 2$ interior sites are channel $B_6$.

For a normal-layer candidate one layer above or below the sheet, the closed footprint intersects the line in two defects over an endpoint column and in three defects over an interior column. For each of the $n$ line columns there are three transverse positions in the normal layer whose footprint sees the line. Therefore each normal layer contributes $3n$ accepted births, and the two normal layers contribute $6n$ accepted births. Endpoint columns contribute 12 normal $B_7$ births in total; interior columns contribute $6(n - 2)$ normal $B_6$ births. Adding the in-plane births gives

$$N_7 = 2 + 12 = 14, \qquad N_6 = (n - 2) + 6(n - 2) = 7n - 14, \qquad N_5 = 0,$$

so $B_0 = N_5 + N_6 + N_7 = 7n$. Active sheet sites adjacent to a one-cell line hole have at most three missing neighbors, hence survive by Theorem 7.3. □

## 13 Non-planar first-response laws

This section records two first-response laws for axis-aligned three-dimensional subfamilies. These are statements about the raw B5–7/S5–9 rule applied to finite binary sets in $\mathbb{Z}^3$ with no boundary wrap-around interaction.

### 13.1 Solid cuboids

Let

$$C_{a,b,c} = \{1, \ldots, a\} \times \{1, \ldots, b\} \times \{1, \ldots, c\}$$

be a solid active cuboid with $a, b, c \geq 4$.

**Theorem 13.1** (Solid cuboid first response). *For a solid $a \times b \times c$ active cuboid with $a, b, c \geq 4$ under the base rule,*

$$A_0 = abc,$$

$$B_0 = 8(a + b + c - 6),$$

$$D_0 = abc - 8,$$

$$A_1 = 8(a + b + c) - 40.$$

*All births occur in channel $B_6$, and all deaths are high-support deaths.*

*Proof.* Inside a solid cuboid, only the eight corners have active neighbor count 7: each corner sees a $2 \times 2 \times 2$ active block including itself, hence seven active neighbors. These eight corner sites survive. Every other active site sees at least ten active neighbors when $a, b, c \geq 4$, so it dies by over-support. Thus $D_0 = abc - 8$.

A birth site outside the cuboid must lie at Chebyshev distance one from the cuboid and see between five and seven active neighbors. The accepted sites are the edge-adjacent exterior strips. Around each of the twelve cuboid edges there are two exterior strips of length equal to the corresponding edge length minus two; the corner neighborhoods contribute the remaining local exterior sites. Summing the edge-direction contributions gives

$$8(a - 2) + 8(b - 2) + 8(c - 2) = 8(a + b + c - 6).$$

Each such exterior birth site sees exactly six active neighbors, so all births are $B_6$. Therefore

$$A_1 = 8 + B_0 = 8 + 8(a + b + c - 6) = 8(a + b + c) - 40.$$

□

### 13.2 Closed surface shells

Let $S_{a,b,c}$ be the closed one-cell-thick surface shell of an $a \times b \times c$ box, with $a, b, c \geq 4$.

**Lemma 13.2** (Closed shell active count). *The number of active sites in the closed surface shell is*

$$A_0 = 2(ab + ac + bc) - 4(a + b + c) + 8.$$

*Proof.* Sum the six faces, giving $2ab + 2ac + 2bc$. Each of the twelve edges has been counted twice, so subtract one copy of each edge: $4(a + b + c) - 12$. The eight corners were then subtracted too far by one each, so add 8. This yields the displayed expression. □

**Theorem 13.3** (Closed surface-shell first response). *For a closed one-cell-thick $a \times b \times c$ surface shell with $a, b, c \geq 4$ under the base rule,*

$$A_0 = 2(ab + ac + bc) - 4(a + b + c) + 8,$$

$$B_0 = 8(a + b + c - 6),$$

$$D_0 = 8(a + b + c - 9),$$

$$A_1 = 2(ab + ac + bc) - 4(a + b + c) + 32.$$

*All births occur in channel $B_6$, and all deaths are high-support deaths.*

*Proof.* The active shell sites split into face-interior sites, edge sites, and corner sites. Face-interior sites have the same local two-dimensional sheet count 8 and survive. Edge sites and corner-adjacent high-support sites see too many active neighbors because two or three shell faces meet in their Moore block. Counting the high-support shell sites along the twelve edges and correcting for the eight corners gives

$$D_0 = 8(a + b + c - 9).$$

The birth layer around a closed shell is the same exterior edge-adjacent collar as in the solid cuboid case. The accepted exterior sites see exactly six active shell neighbors, so births are all $B_6$, and

$$B_0 = 8(a + b + c - 6).$$

Subtracting deaths and adding births to the initial shell count gives

$$A_1 = A_0 - D_0 + B_0 = 2(ab + ac + bc) - 4(a + b + c) + 32.$$

□

**Remark 13.4** (Piecewise non-planar families)**.** *Open ports, slits, tunnels, and attached walls introduce case distinctions depending on port width, contact with edges, and overlapping Moore footprints. They are not included in Theorem 13.1 or Theorem 13.3.*

# 14 Channel-resolved PooleQ/P transform

The raw update collapses all accepted birth and survival events into a single live/dead output. The channel-resolved transform preserves the local event type.

**Definition 14.1** (PooleQ/P local feature vector)**.** *For a binary field $x$, define*

$$\Phi_i(x) = (B_5, B_6, B_7, S_5, S_6, S_7, S_8, S_9, O_{10+}, \psi)_i,$$

*where*

$$O_{10+}(i; x) = x(i)\mathbf{1}\{N_i(x) \geq 10\}$$

*and $\psi_i$ is the support strain defined earlier.*

**Remark 14.2** (Channel observability)**.** *The vector $\Phi_i$ is richer than the next binary state $P(x)_i$. If only $P(x)_i$ is observed, then $B_5, B_6, B_7$ are collapsed into one birth event and the survival channels are likewise collapsed. Channel-resolved use therefore assumes direct local feature extraction, typed readout, or a spatial layout that preserves channel identity.*

# 15 Probabilistic p-bit form

A p-bit is a stochastic binary variable with a controlled probability of being 1; such systems are used in probabilistic computing models [3, 4]. The following formulas are classical probability identities on local Bernoulli fields.

Let $X_j \sim \text{Bernoulli}(p_j)$ independently over $j \in \Lambda_L$. Since the center site is not included in $N_i$, the center variable $X_i$ is independent of $N_i$ under the independent-site model.

**Definition 15.1** (Local generating function)**.** *For a fixed site $i$, define*

$$G_i(z) = \prod_{j \in \mathcal{N}_{26}(i)} (1 - p_j + p_j z).$$

*Then*

$$\Pr[N_i(X) = k] = [z^k] G_i(z).$$

**Proposition 15.2** (Expected Poole channels)**.** *Under the independent Bernoulli model,*

$$\mathbb{E}[B_k(i; X)] = (1 - p_i)[z^k] G_i(z),$$

$$\mathbb{E}[S_k(i; X)] = p_i [z^k] G_i(z).$$

*Therefore the expected Poole activation is*

$$q_i := \mathbb{E}[P(X)_i] = (1 - p_i) \sum_{k=5}^{7} [z^k] G_i(z) + p_i \sum_{k=5}^{9} [z^k] G_i(z).$$

*Equivalently,*

$$q_i = C_i^{5:7} + p_i C_i^{8:9},$$

*where $C_i^{a:b} = \sum_{k=a}^{b} [z^k] G_i(z)$.*

*Proof.* The generating function is the standard product for a sum of independent, non-identically distributed Bernoulli variables. Since $X_i$ is independent of the neighbor count, multiplying by $1 - p_i$ or $p_i$ gives the birth and survival expectations. Summing the accepted channels gives the activation formula. The final expression follows from $C_i^{5:9} = C_i^{5:7} + C_i^{8:9}$. □

**Proposition 15.3** (Differentiability)**.** *For $\ell \in \mathcal{N}_{26}(i)$,*

$$\frac{\partial}{\partial p_\ell} [z^k] G_i(z) = [z^k] \left( (z - 1) \prod_{j \in \mathcal{N}_{26}(i), j \neq \ell} (1 - p_j + p_j z) \right).$$

*For the center site,*

$$\frac{\partial q_i}{\partial p_i} = C_i^{8:9}.$$

*Proof.* Differentiate the product defining $G_i(z)$. The center derivative follows from $q_i = C_i^{5:7} + p_i C_i^{8:9}$ and the fact that the center probability is not a factor in the neighbor generating function. □

## 16 Correlation residue

Suppose $M$ measured binary fields $X^{(1)}, \ldots, X^{(M)}$ are available. Define empirical channel rates

$$\widehat{B}_k(i) = \frac{1}{M} \sum_{m=1}^{M} B_k(i; X^{(m)}), \qquad \widehat{S}_k(i) = \frac{1}{M} \sum_{m=1}^{M} S_k(i; X^{(m)}),$$

and empirical marginals

$$\widehat{p}_i = \frac{1}{M} \sum_{m=1}^{M} X_i^{(m)}.$$

Using $\widehat{p}_j$ in the independent formula gives an independent prediction for channel rates. A channel-wise residue vector may be defined as

$$R_i = (\widehat{B}_5 - \mathbb{E}_{ind}B_5, \widehat{B}_6 - \mathbb{E}_{ind}B_6, \widehat{B}_7 - \mathbb{E}_{ind}B_7, \widehat{S}_8 - \mathbb{E}_{ind}S_8, \widehat{S}_9 - \mathbb{E}_{ind}S_9)_i.$$

A scalar local-dependence statistic is

$$C_P = \sum_{i \in \Lambda_L} \|R_i\|_2^2.$$

This statistic detects departure from the independent-site local model. It does not by itself identify a physical noise source.

# 17 Defect cardinality logic

Around a stable Poole sheet, the channel-resolved birth rule becomes exact-cardinality logic.

Let $D$ be a set of missing cells in a stable sheet. For a normal-layer candidate above or below sheet location $u$, the active neighbor count is

$$N = 9 - r_D(u).$$

**Proposition 17.1** (Channel-cardinality equivalence)**.** *For a normal-layer candidate adjacent to a defective sheet,*

$$B_k = 1 \quad \Longleftrightarrow \quad r_D(u) = 9 - k, \qquad k \in \{5, 6, 7\}.$$

*Thus*

$$B_7 \Longleftrightarrow r_D = 2, \quad B_6 \Longleftrightarrow r_D = 3, \quad B_5 \Longleftrightarrow r_D = 4.$$

*Proof.* Normal-layer birth is accepted when $5 \leq N \leq 7$, and its channel is the value of $N$. Since $N = 9 - r_D$, channel $B_k$ is equivalent to $9 - r_D = k$. □

## 17.1 Biased exact-cardinality detectors

Let $A_1, \ldots, A_n \in \{0, 1\}$ be input defects placed at selected footprint positions, and let $b$ be the number of fixed bias defects in the same $3 \times 3$ footprint. If

$$s = \sum_{j=1}^{n} A_j, \qquad r = s + b,$$

then

$$B_7 = \mathbf{1}\{s = 2 - b\}, \quad B_6 = \mathbf{1}\{s = 3 - b\}, \quad B_5 = \mathbf{1}\{s = 4 - b\}.$$

This is a local exact-count circuit.

## 17.2 Boolean gates

Using the collapsed raw birth event

$$B_{raw} = B_5 \vee B_6 \vee B_7,$$

one obtains small biased-footprint Boolean gates. For two inputs $A, B$ and bias $b$, the raw birth condition is

$$5 \leq 9 - (A + B + b) \leq 7,$$

which is equivalent to

$$2 \le A + B + b \le 4.$$

Choosing the bias changes the accepted input cardinalities. Channel-resolved readout gives exact cardinality; collapsed readout gives thresholded cardinality.

| Bias $b$ | Accepted input sums | Collapsed raw function |
|---|---|---|
| 0 | 2 | AND of two active defects |
| 1 | $1, 2$ | OR of two active defects |
| 2 | $0, 1, 2$ | constant true for two inputs |
| 3 | $0, 1$ | NAND of two active defects |
| 4 | 0 | NOR of two active defects |

Complemented output can also be obtained by reading absence of the raw birth event in the same local layout.

## 18 Adder readouts

For two inputs with bias $b = 1$, the channel identity directly distinguishes the three possible input sums: no birth means $A + B = 0$, channel $B_7$ means $A + B = 1$, and channel $B_6$ means $A + B = 2$. Thus a half-adder can be read as

$$\text{sum} = \mathbf{1}\{B_7\}, \qquad \text{carry} = \mathbf{1}\{B_6\}.$$

| $A$ | $B$ | channel at $b = 1$ | sum | carry |
|---|---|---|---|---|
| 0 | 0 | none | 0 | 0 |
| 1 | 0 | $B_7$ | 1 | 0 |
| 0 | 1 | $B_7$ | 1 | 0 |
| 1 | 1 | $B_6$ | 0 | 1 |

For three inputs with bias $b = 1$, no birth means $A + B + C = 0$, $B_7$ means $A + B + C = 1$, $B_6$ means $A + B + C = 2$, and $B_5$ means $A + B + C = 3$. A full-adder readout is therefore

$$\text{sum} = \mathbf{1}\{B_7\} + \mathbf{1}\{B_5\}, \qquad \text{carry} = \mathbf{1}\{B_6\} + \mathbf{1}\{B_5\}.$$

| Input sum $s$ | channel at $b = 1$ | sum bit $s$ mod 2 | carry bit $\mathbf{1}\{s \ge 2\}$ |
|---|---|---|---|
| 0 | none | 0 | 0 |
| 1 | $B_7$ | 1 | 0 |
| 2 | $B_6$ | 0 | 1 |
| 3 | $B_5$ | 1 | 1 |

**Remark 18.1** (Readout boundary)**.** *The adder construction is a channel-readout identity, not a claim that an uninstrumented one-bit next-state automaton preserves every intermediate channel. Raw next-state observation alone loses channel identity.*

## 19 Measured qubit syndrome boundary

Surface-code and syndrome-decoding contexts provide measured classical binary fields after quantum measurement [5, 6, 7]. The PooleQ/P transform may be applied to such already-measured fields as a classical local feature map. It does not copy, clone, or reconstruct an unmeasured quantum state. Its mathematical input is a binary field, a probability field, or a sequence of measured syndrome bits.

## 20 Signed local interface model

**Definition 20.1** (Signed lattice). *A signed configuration is a map*

$$s : \Lambda_L \to \{-1, 0, +1\}.$$

*The sets of positive and negative active sites are*

$$P_s = \{i : s(i) = +1\}, \qquad M_s = \{i : s(i) = -1\}.$$

*Let $N_i^+$ and $N_i^-$ be the counts of positive and negative neighbors in the Moore neighborhood.*

**Definition 20.2** (Signed support update). *Fix birth and survival windows $B[a, b]$ and $S[c, d]$, and an opposite-support collapse threshold $k$. A positive candidate is born at an empty site when $N_i^+ \in [a, b]$ and is retained at a positive site when $N_i^+ \in [c, d]$; it is removed if $N_i^- \geq k$. The negative candidate rule is the sign-reversed analogue. If both signs are candidates at the same site, the site is assigned $0$.*

**Definition 20.3** (Interface support and balance). *Define the local signed overlap proxy*

$$q_i = \min(N_i^+ + \mathbf{1}\{s(i) = +1\},\ N_i^- + \mathbf{1}\{s(i) = -1\}).$$

*For a region $\Omega$, define*

$$Q = \sum_{i \in \Omega} q_i \mathbf{1}\{q_i \geq q_0\}, \qquad Q_\rho = \frac{Q}{|\Omega|}.$$

*Let*

$$\epsilon_\pm = \frac{|N_+ - N_-|}{N_+ + N_- + \varepsilon}$$

*be signed imbalance. If an axis is chosen, let $\sigma_y$ be the support-weighted standard deviation of interface support along that axis.*

**Definition 20.4** (Signed membrane descriptor). *The signed membrane quality descriptor is*

$$M_P = \frac{Q_\rho}{1 + \sigma_y}(1 - \epsilon_\pm).$$

*It is a compact morphology descriptor: interface density multiplied by thinness and signed balance. It is not a physical energy law, thrust law, or propulsion law.*

## 21 The defect-channel bridge

The planar defect calculus and the PooleQ/P channel transform are two encodings of the same first-step local count information.

**Theorem 21.1** (Defect-channel bridge)**.** *Let $x_D$ be a defective stable sheet. In the first update:*

1. *for an active sheet site $u \notin D$, the survival channel is*

$$S_{8-q_D(u)};$$

2. *for an in-plane defect site $u \in D$, the birth channel is*

$$B_{8-q_D(u)};$$

3. *for a normal-layer site $(u, c \pm 1)$, the birth channel is*

$$B_{9-r_D(u)}.$$

*Each channel is active only when its index lies in the accepted channel set.*

*Proof.* Theorem 7.3 gives the raw neighbor counts. Active sheet sites have $N = 8 - q_D(u)$, in-plane defect sites have $N = 8 - q_D(u)$, and normal-layer candidates have $N = 9 - r_D(u)$. The channel label is exactly the raw neighbor count for the corresponding birth or survival event. □

**Corollary 21.2** (Spectrometer interpretation)**.** *For first-step planar defects, the pair of planar functions $(q_D, r_D)$ can be recovered from the typed channel labels on the sheet and adjacent normal layers, within the accepted event region. Conversely, the channel labels are determined by $(q_D, r_D)$.*

## 22 Many-to-one collapse of raw next-state observation

**Proposition 22.1** (Channel identity is lost under raw next-state observation)**.** *There exist distinct local neighborhoods with different channel labels that produce the same next binary state.*

*Proof.* For an inactive center site, neighborhoods with $N = 5$, $N = 6$, and $N = 7$ all produce $P(x)_i = 1$, but their channel labels are $B_5, B_6, B_7$. Therefore the raw next-state bit does not determine the channel. The same holds for active center sites with $N = 5, 6, 7, 8, 9$, all of which survive but have different survival channels. □

## 23 Claim boundary and open problems

The established mathematical layer consists of the finite local theorems and formulas above. The following boundaries remain.

1. General multi-step planar relaxation is not closed by the first-step selector. The isolated pore-gas subfamily is fixed, but arbitrary cracks, rectangles, and clustered masks may enter longer cascades.

2. Open-port, slit, tunnel, and attached-wall non-planar families require piecewise local classifications beyond the solid cuboid and closed shell laws.

3. PM$\varphi$ is a tagged first-step response model and must not be confused with the raw base rule.
4. Q/P channel use requires channel-resolved feature extraction. Raw next-state observation alone is many-to-one.
5. The signed interface descriptor $M_P$ is a morphology score, not a physical performance law.
6. Continuum or physical interpretations require separate derivations and external validation.

## 24 Conclusion

The base Poole rule is a compact hysteretic support law whose local threshold structure selects one-cell sheets, rejects lower-dimensional and filled bulk flats, and gives an exact first-step calculus for planar defects. The planar selector reduces a three-dimensional local update near a defective sheet to two two-dimensional footprint counts, $q_D$ and $r_D$. Rectangle defects produce closed first-step count, channel, and work formulas. Solid cuboids and closed surface shells provide two non-planar first-response families. The channel-resolved PooleQ/P transform preserves the hidden birth, survival, high-support, and strain channels of the same rule and admits classical probability-layer identities. The signed model extends the support-window logic to opposite-signed local populations while keeping its morphology scores explicitly nonphysical. The central bridge theorem shows that the Defect Calculus selector and the Q/P channel spectrum are the same first-step local information expressed in different coordinates.

## A Theorem index

This appendix lists the main mathematical statements by lane. It is included to make the submission navigable without referring to any outside manuscript.

| Statement | Lane | Content |
|---|---|---|
| Compact-rule equivalence | theorem | The compact formula is exactly B5–7/S5–9. |
| Finite propagation | theorem | State at a site after $t$ steps depends only on the radius-$t$ Moore ball. |
| Eventual periodicity | theorem | Every finite-lattice orbit is eventually periodic. |
| Non-monotonicity | theorem | Adding active support can destroy inactive birth eligibility. |
| Non-reversibility | theorem | All-zero and all-one configurations have the same image. |
| Surface selector | theorem | Coordinate sheets survive while ideal points, lines, and filled blocks do not. |
| Stable plane | theorem | A full coordinate plane is a fixed point. |
| Planar defect selector | theorem | First response near a defective plane is determined by $q_D$ and $r_D$. |
| Pore-gas fixed subfamily | theorem | $r_D \le 1$ everywhere implies fixedness. |
| Rectangle laws | theorem/formula | Rectangles have closed first-step count, channel, and work identities. |
| Solid cuboid law | theorem/formula | Solid boxes have closed first-response formulas for $a, b, c \ge 4$. |
| Closed surface-shell law | theorem/formula | Closed one-cell shell boxes have closed first-response formulas for $a, b, c \ge 4$. |
| Poisson-binomial layer | theorem/formula | Expected Q/P channels have local generating-function expressions. |
| Defect-channel bridge | theorem | $q_D$ and $r_D$ determine first-step Q/P channel labels. |
| Channel collapse | theorem | Raw next-state bits do not determine channel identity. |

# B Planar operator form

The planar selector can be written as four set-valued operators on the defect mask $D \subset \mathbb{T}_L$:

$$\mathcal{D}(D) = \{u \notin D : q_D(u) \ge 4\},$$

$$\mathcal{B}_{\|}(D) = \{u \in D : 1 \le q_D(u) \le 3\},$$
$$\mathcal{B}_{+}(D) = \{(u, c+1) : 2 \le r_D(u) \le 4\},$$
$$\mathcal{B}_{-}(D) = \{(u, c-1) : 2 \le r_D(u) \le 4\}.$$

Here $\mathcal{D}$ is the active sheet death set, $\mathcal{B}_{\|}$ is the in-plane birth set, and $\mathcal{B}_{\pm}$ are the two normal-layer birth sets. The first-step event set is exactly

$$\mathcal{E}(D) = \mathcal{D}(D) \cup \mathcal{B}_{\|}(D) \cup \mathcal{B}_{+}(D) \cup \mathcal{B}_{-}(D).$$

No site outside these four sets can change at the first update.

The channel-decomposed normal-layer operators are

$$\mathcal{B}_{\pm,k}(D) = \{(u, c \pm 1) : 9 - r_D(u) = k\}, \qquad k \in \{5, 6, 7\}.$$

The in-plane channel operators are

$$\mathcal{B}_{\parallel,k}(D) = \{u \in D : 8 - q_D(u) = k\}, \qquad k \in \{5, 6, 7\}.$$

The sheet survival channel operators are

$$\mathcal{S}_k(D) = \{u \notin D : 8 - q_D(u) = k\}, \qquad k \in \{5, 6, 7, 8, 9\}.$$

# C Detailed local proof of the no-event sufficient condition

If $r_D(u) \leq 1$ for all $u$, then no two defect cells occur in any closed $3 \times 3$ planar footprint. Consequently each defect site has no defect neighbor in its open footprint, so $q_D = 0$ on $D$ and in-plane rebirth is impossible. Each active site adjacent to a pore sees at most one defect in its open footprint, so its support is at least 7 and at most 8, hence it survives. Every normal-layer candidate sees $N = 9 - r_D \geq 8$, which is outside the birth window $5, 6, 7$. All sites at distance at least two from the plane see 0 active neighbors. Therefore every site retains its initial value.

# D Rectangle footprint classification

For a rectangular defect with $a, b \geq 2$, the accepted birth sites are classified by the value of $N$, equivalently by $q_D$ or $r_D$. The stable-plane geometry implies that births can occur only in the original sheet plane or in the two adjacent normal layers. The accepted in-plane births lie on the non-isolated part of the rectangle boundary. The accepted normal births lie in the two closed-footprint collars for which the normal support $9 - r_D$ is 5, 6, or 7.

The channel spectrum

$$(N_5, N_6, N_7) = (12, 4a + 4b - 16, 16)$$

can be read as follows. The twelve $B_5$ sites are the highest-defect accepted footprint sites. The sixteen $B_7$ sites are the shallow accepted corner-edge sites. The remaining side contribution is linear in the side lengths and equals $4a + 4b - 16$. The sum is

$$12 + (4a + 4b - 16) + 16 = 4a + 4b + 12.$$

This classification is independent of the absolute position of the rectangle because the rule is translation-equivariant.

# E Rectangle reconstruction under a rectangle prior

Under a rectangle prior with $a, b \geq 2$, the raw first-step birth count gives

$$a + b = \frac{B_0 - 12}{4}.$$

The channel $B_6$ gives the same quantity by

$$a + b = \frac{N_6 + 16}{4}.$$

Thus the first response determines the semiperimeter. It does not by itself determine the unordered pair $\{a, b\}$ unless an additional aspect observable, orientation prior, or second independent measurement is supplied. Any stronger reconstruction claim must state the extra prior explicitly.

## F Dynamic programming form for the probability layer

For numerical or symbolic evaluation of the Poisson-binomial coefficients, the generating function can be built by the recurrence

$$F_0(0) = 1, \qquad F_0(k) = 0 \quad (k \neq 0),$$

$$F_{m+1}(k) = (1 - p_{j_m})F_m(k) + p_{j_m}F_m(k-1),$$

where $j_1, \dots, j_{26}$ enumerate the Moore neighbors of $i$. Then

$$F_{26}(k) = [z^k]G_i(z) = \Pr[N_i(X) = k].$$

Only coefficients $0 \leq k \leq 26$ are needed. The expected activation reduces to the two local bands $5:7$ and $8:9$:

$$q_i = C_i^{5:7} + p_i C_i^{8:9}.$$

This formula isolates the hysteretic part: the center probability $p_i$ only gates the upper memory band.

## G Model dictionary

**PM0.** The raw base Poole map B5–7/S5–9.

**PM$\varphi$.** A tagged first-step rectangle response model in which the raw $B_7$ group is suppressed. It is not the raw map.

$q_D$**.** Open planar defect count in the eight-neighborhood of a sheet site.

$r_D$**.** Closed planar defect count in the $3 \times 3$ footprint of a sheet site.

$B_k$**.** Birth channel with raw neighbor count $k$.

$S_k$**.** Survival channel with raw neighbor count $k$.

$\psi$**.** Signed support strain, separating under-support and over-support.

$M_P$**.** Signed membrane morphology descriptor, not a physical performance law.

## H Open theorem problems

1. Prove or refute the full no-event converse: localized planar masks with no first-step events are exactly isolated pore gases.
2. Derive closed multi-step relaxation laws for selected planar families beyond the fixed pore-gas case.
3. Complete a piecewise first-response classification for boxes with ports, slits, tunnels, and attached walls.
4. Determine which channel-resolved local circuits can be embedded without interfering footprints.
5. Characterize signed-interface fixed points and cycles under specified collapse thresholds.
6. Derive any continuum limit only after specifying scaling, observables, topology, and convergence mode.

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